

UCT602: INFORMATION SECURITY

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Course Objectives: Learn the principles of information security to protect the confidentiality, integrity, and availability of information. Discuss the modes of threats and attacks on information systems, threat mitigation, cryptography, user identification and authentication, access control, privacy laws, and more.

Overview of Security Parameters: Confidentiality, integrity and availability; Security violation and threats; Security policy and procedure; Assumptions and Trust; Security Assurance, Implementation and Operational Issues; Security Life Cycle.

Access Control Models: Discretionary, mandatory, roll-based and task-based models, unified models, access control algebra, temporal and spatio-temporal models.

Security Policies: Confidentiality policies, integrity policies, hybrid policies, non-interference and policy composition, international standards.

Systems Design: Design principles, representing identity, control of access and information flow, confinement problem. Assurance: Building systems with assurance, formal methods, evaluating systems.

Logic-based System: Malicious logic, vulnerability analysis, auditing, intrusion detection. Applications: Network security, operating system security, user security, program security.

Special Topics: Data privacy, introduction to digital forensics, enterprise security specification.

Operating Systems Security: Security Architecture, Analysis of Security in Linux/Windows.

Database Security: Security Architecture, Enterprise security, Database auditing.

Laboratory Work:

1. Analysis of security in Unix/Linux.
2. Administration of users, password policies, privileges and roles

Course Learning Outcomes (CLOs) / Course Objectives (COs):

Upon completion of this course, students will be able to:

1. Understand the technical, social & political aspects of Information and its applications.
2. Analyze the vulnerabilities and threats with respect to various types of attacks.
3. Understand the access control systems and their applicability in wide range of applications.

4. Recognize the attacks and defense patterns during information processing and management.

Text Books:

1. Security Engineering, Ross Anderson.
2. Computer Security: Art and Science, M. Bishop, Pearson Education.
3. Information Security: Principles and Practice, M. Stamp.

Reference Books:

3. Security in Computing, C.P. Pfleeger, S.L. Pfleeger, J. Margulies.
4. Secure Programming HOWTO, David Wheeler.
5. Browser Security Handbook, Michael Zalewski.
6. Handbook of Database Security, M. Gertz, S. Jajodia.